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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

IS3321-1 – BLOCKCHAIN TECHNOLOGY

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What connects cryptocurrency to blockchain technology?**
A) Cryptocurrencies use blockchain as a ledger for transactions
B) Blockchain is used to store private keys
C) Cryptocurrencies encrypt the blockchain
D) Blockchain mines cryptocurrency automatically
2. **What is a permissioned blockchain?**
A) A blockchain where anyone can participate without restrictions.
B) A blockchain accessible only to authorized participants.
C) A blockchain that uses a proof-of-stake mechanism.
D) A blockchain designed for mining Bitcoin
3. **Which consensus mechanism is used by Bitcoin?**
A) Proof of Stake (PoS)
B) Delegated Proof of Stake (DPoS)
C) Proof of Work (PoW)
D) Byzantine Fault Tolerance (BFT)
4. **What is the primary advantage of blockchain security?**
A) High encryption strength
B) Low cost of implementation
C) Centralized control of data
D) Decentralization and immutability
5. **Which of the following is NOT a feature of a private blockchain?**
A) Faster transaction processing
B) Public participation
C) Restricted access
D) Higher level of control
6. **What does the term "cryptographic hash" refer to in blockchain?**
A) A unique fixed-length output generated from input data
B) A public key used to encrypt data
C) A password to access blockchain
D) The encryption of private keys
7. **What is the main role of miners in a blockchain?**
A) To store data on the blockchain
B) To decrypt encrypted blocks
C) To create cryptocurrency wallets
D) To validate and add transactions to the blockchain
8. **Which of the following is a common use case for permissioned blockchains?**
A) Anonymous cryptocurrency transactions
B) Decentralized gaming platforms
C) Supply chain management and enterprise applications
D) Crowdfunding platforms
9. **What is the "view change" process in Byzantine Fault Tolerance?**
A) A change in network hash rate
B) Switching from one leader to another in case of failure
C) Altering the blockchain protocol
D) Adjusting mining difficulty

10. **What is the primary challenge addressed by the Byzantine Generals Problem?**
 A) Consensus with faulty or malicious nodes B) Scalability in distributed systems
 C) Network latency D) High transaction fees
11. **How many Byzantine faults can a system tolerate if there are $3f + 1$ nodes?**
 A) f faults B) $3f$ faults
 C) $2f$ faults D) $4f$ faults
12. **What is the purpose of a quorum in Paxos?**
 A) To increase the number of miners in the network B) To validate transactions quickly
 C) To ensure a majority agreement in distributed consensus D) To reduce energy consumption
13. **What is the significance of message ordering in distributed consensus systems?**
 A) It simplifies cryptographic calculations B) It reduces transaction costs
 C) It accelerates mining speed D) It ensures consistent state replication across nodes
14. **What is the purpose of decentralized identity management on blockchain?**
 A) To allow individuals full control over their personal identity data B) To eliminate the need for identity verification
 C) To centralize identity verification processes D) To encrypt user credentials permanently
15. **What is Hyperledger Fabric?**
 A) A public blockchain for cryptocurrency transactions B) A blockchain exclusively for Ethereum smart contracts
 C) A permissioned blockchain platform designed for enterprise use D) A blockchain exclusively for Ethereum smart contracts
16. **What is the key advantage of using smart contracts on Ethereum?**
 A) Complete automation of agreements without intermediaries B) Increased mining rewards
 C) Anonymous identity verification D) Centralized processing
17. **What is the "Membership and Access Control" feature in Hyperledger Fabric?**
 A) A mechanism to prevent transaction tampering B) A way to define and enforce roles and permissions within the network
 C) A feature for creating mining pools. D) A system for managing public key encryption
18. **What is "Proof of Stake" in Ethereum 2.0?**
 A) B) A mechanism to mine tokens faster B) A way to ensure anonymous participation
 C) A consensus mechanism where validators are chosen based on their stake in the network. D) A protocol to eliminate transaction fees
19. **What is the role of "Channels" in Hyperledger Fabric?**
 A) To group transactions into blocks B) To store smart contracts
 C) To validate cryptocurrency transactions D) To create private, isolated communication pathways for subsets of participants
20. **What is the role of the "Orderer" in Hyperledger Fabric?**
 A) To endorse transactions before execution B) To collect transactions and create blocks in a specific sequence
 C) To mine cryptocurrency tokens D) To encrypt channel communications

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain Payments and Double Spending and also interpret how double spending is handled using blockchain technology.	8	L*2	1	1
	b) Explain provenance tracking and IPFS application of blockchain technology.	8	L2	2	1
2.	a) What is distributed consensus? Explain the working of PoW consensus algorithm in bitcoin mining,	8	L2	2	1

b)	Explain joining and transaction flooding in bitcoin P2P network.	8	L2	2	1
3. a)	What is monopoly problem? Explain how monopoly problem and power consumption problem can be handled in blockchain.	8	L2	2	1
b)	Compare and contrast the following i) Centralized, Decentralized and Distributed system ii) POW, POS and POB	8	L2	1	1
Unit – II					
4. a)	Why 3f+1 replicas are required to ensure safety in an asynchronous system when there are f faulty nodes ?	6	L2	3	1
b)	What is smart contract? Explain the design issues of smart contracts in permissioned model.	6	L2	3	1
c)	List the differences between synchronous and asynchronous environment.	4	L1	3	1
5. a)	What is distributed consensus? Outline the process of how state machine replication executes contract at subset of nodes.	8	L2	3	1
b)	Illustrate cross border payment using blockchain.	8	L2	4	1
6. a)	Explain view change process in PBFT algorithm.	8	L2	3	1
b)	Illustrate the role of blockchain in food safety.	8	L2	4	1
Unit – III					
7. a)	Outline the architecture of Hyperledger Fabric and list the steps involved in transaction validation.	8	L2	5	1
b)	What is Ethereum virtual machine? Explain different account types, gas and transactions in Ethereum blockchain.	8	L2	5	1
8. a)	Explain Membership and Access control architecture of Fabric.	8	L2	5	1
b)	Explain the identities and policies required at every stage of fabric network.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
